



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE NAME: Theory of Computation

COURSE CODE: CSA1310

COURSE OUTCOME COVERED

CO1: *Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems (BL:3)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Duration : 1 Hour

Technical Presentation

Code of Conduct:

I T. Divya (192411120) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

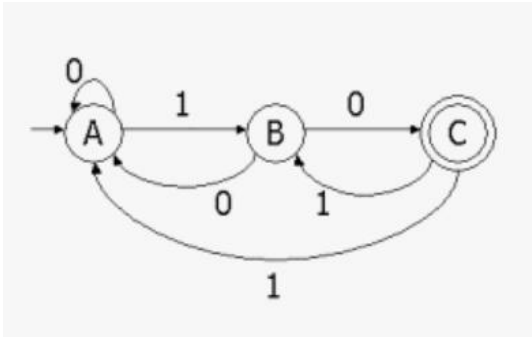
Signature of the student:



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1. How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language L



Automaton Simulator: DFA NFA PDA Examples

Test / Debug:

Rejected

Bulk Testing

Accept (one per line):

0
00
100

Reject (one per line):

1
0 1

```
graph LR; start((start)) -- 0 --> start; start -- 1 --> B((B)); B -- 1 --> B; B -- 0 --> start; B -- 0 --> C(((C))); C -- 1 --> B; C -- 1 --> start;
```

Test Results:

Accept: 0 -- Pass
Accept: 00 -- Pass
Accept: 100 -- Pass
Reject: 1 -- Pass
Reject: 0 1 -- Pass

Kyle Dickerson -- kyle.dickerson@gmail.com -- On GitHub Tools: jQuery jsPlumb Icons: FatCow

34°C Mostly sunny

Search

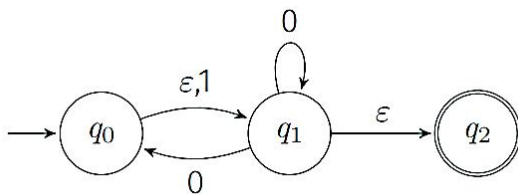
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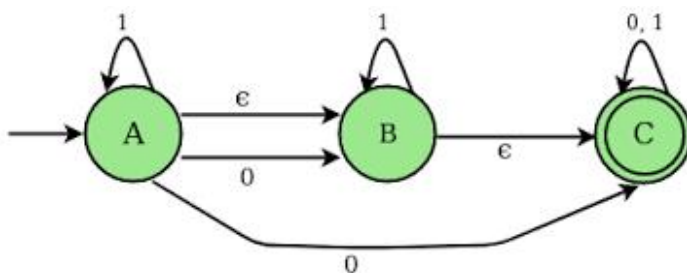
2. How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integers n , the statement given below is true.

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

3. What role do epsilon (ϵ) transitions play in finite automata, and how can ϵ -NFA be converted into an equivalent NFA without epsilon transitions? Construct an equivalent NFA without ϵ -transitions that accepts the same language. Draw its transition table and mark the initial and final states



4. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and construct an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states





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RUBRICS FOR CALCULATION

Criteria	Weight age	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50

